

Clumpiness calculation

[Publication](#), [GitHub](#)

$$C(L) = \frac{1}{n} \left(\prod_{i=1}^n \frac{x}{y^i} \right)^{\frac{1}{n}}$$

n - Number of unique labels.

y_i - Frequency of label i in the tree

$$y_i = \frac{|L_i|}{|T|}$$

L_i - Number of leaves with label i

T - total number of relevant leaves

x - the weighted “average” of inner vertices in the tree

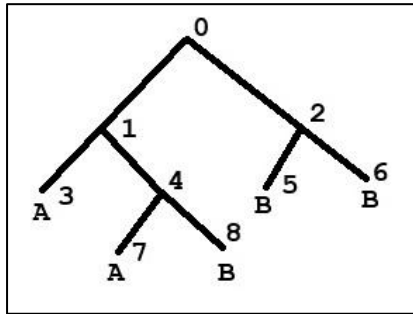
$$x = \frac{1}{|I|} \sum_{v \in I} \delta(v) \omega(v)$$

$|I|$ - Total number of inner (non-leaf) nodes, excluding the root node.

$\delta(v)$ - A viability indicator. It equals 1 if the inner vertex ‘ v ’ has at least one of each label in its downstream descendant leaves. If it is missing any label among its descendants, it equals to 0.

$\omega(v)$ - A weight applied to a viable vertex based on how many branching steps away its descendant leaves are. Every time a branch splits, the weight of the path is divided by the number of children (e.g., multiplied by ‘1/2’ for a binary tree).

Example – Clumpiness Calculation



node_id	parent_node_id	label
0	nan	nan
1	0	nan
2	0	nan
3	1	a
4	1	nan
5	2	b
6	2	b
7	4	a
8	4	b

$$C(L) = \frac{1}{n} \left(\prod_{i=1}^n \frac{x_i}{y_i} \right)^{\frac{1}{n}}$$

$n = 2$ (number of unique labels)

$$y_A = \frac{|L_i|}{|T|} = \frac{2}{5} = 0.4$$

$$y_B = \frac{|L_i|}{|T|} = \frac{3}{5} = 0.6$$

$$\delta(1) = 1$$

$$\omega(1) = \frac{1}{2} + \left(\frac{1}{2} * \frac{1}{2} \right) + \left(\frac{1}{2} * \frac{1}{2} \right) = \frac{2}{4} + \frac{1}{4} + \frac{1}{4} = 1$$

$$\delta(2) = 0$$

$$\omega(2) = \frac{1}{2} + \frac{1}{2} = 1$$

$$\delta(4) = 1$$

$$\omega(4) = \frac{1}{2} + \frac{1}{2} = 1$$

$$x = \frac{1}{|I|} \sum_{v \in I} \delta(v) \omega(v) = \frac{1}{3} [(1 * 1) + (1 * 0) + (1 * 1)] = \frac{1}{3} * \frac{2}{1} = \frac{2}{3} \approx 0.667$$

$$C(A, B) = \frac{1}{n} \left(\prod_{i=1}^n \frac{x_i}{y_i} \right)^{\frac{1}{n}} = \frac{1}{2} \left(\frac{\frac{2}{5}}{\frac{3}{5}} * \frac{\frac{2}{5}}{\frac{3}{5}} \right)^{\frac{1}{2}} = \frac{1}{2} * \left(\frac{5}{3} * \frac{10}{9} \right)^{\frac{1}{2}} \cong 0.680$$

Find clumpiness results output:

property1	property2	value
a	a	0.319586
a	b	0.680414
b	a	0.680414
b	b	0.319586